GATE ME 90 Days Challenge

Day 78/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Full Syllabus	Full-Length Mock 2	21 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on a mixed full-syllabus warm-up for the second full-length mock: ages, subject–verb agreement, limits, probability densities, truss joints, cantilever bending stress, compound gear trains, damped vibration, bearing life, jets on moving plates, thermocouple time constants, wet steam, Rankine efficiency, shear angle and LP corner points. Try them in **60 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 11 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 65 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

The present ages of A and B are in the ratio 5 : 7. Eight years from now the ratio will be 7 : 9. The present age of A (in years) is _____

Q2 MCQ

Select the sentence that is grammatically correct.

(A) Each of the test specimens have been numbered.

(B) Neither of the welds are defective.

(C) Each of the test specimens has been numbered.

(D) The number of rejected parts have increased this month.

Q3 MCQ

The value of the limit of $(1 + 3/x)^{2x}$ as $x \to \infty$ is

(A) 1

(B) $e^{3/2}$

(C) $e^{2/3}$

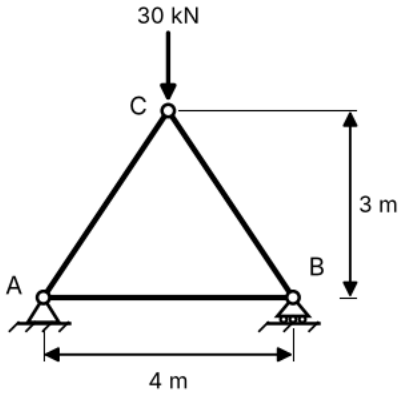
(D) e^{6}

Q4 NAT

A random variable X has the probability density function $f(x) = kx(2 - x)$ for $0 \le x \le 2$ and zero elsewhere. The probability $P(X < 0.5)$ (round off to three decimal places) is _____

Q5 MCQ

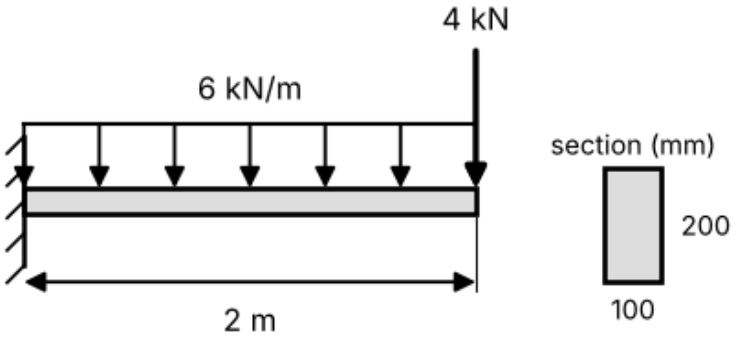
The triangular truss shown (span 4 m, height 3 m, apex C above the mid-point of AB) has a pin at A and a roller at B, and carries a vertical load of 30 kN at C. The forces in member AC and member AB are



(A) 15.0 kN compression and 20.0 kN tension
(B) 18.0 kN compression and 10.0 kN tension
(C) 18.0 kN tension and 10.0 kN compression
(D) 10.0 kN compression and 18.0 kN tension

Q6 NAT

The cantilever shown, 2 m long, carries a uniformly distributed load of 6 kN/m over its whole length and a point load of 4 kN at its free end. Its cross-section is a rectangle 100 mm wide and 200 mm deep. The maximum bending stress (in MPa) is _____



Q7 MCQ

In a compound gear train, gear A (20 teeth) on the input shaft drives gear B (60 teeth). Gear C (25 teeth) is keyed to the same shaft as B and drives gear D (75 teeth) on the output shaft. All gears are external. If the input shaft turns at 1440 rpm, the output shaft turns at

(A) 160 rpm in the same direction as the input
(B) 160 rpm opposite to the input
(C) 480 rpm in the same direction as the input
(D) 12 960 rpm opposite to the input

Q8 NAT

A spring–mass–damper system has m = 4 kg, k = 1600 N/m and c = 48 N·s/m. Its damped natural frequency (in rad/s, round off to two decimal places) is _____

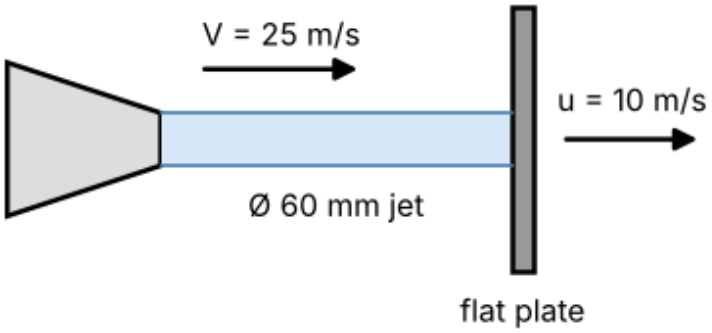
Q9 MCQ

The radial load on a cylindrical roller bearing is increased by 20 % while the speed stays the same. Its rating life in hours becomes about

(A) 58 % of the original
(B) 69 % of the original
(C) 54 % of the original
(D) 83 % of the original

Q10 NAT

A water jet of 60 mm diameter issues at 25 m/s and strikes normally a flat plate that moves away from the nozzle at 10 m/s in the direction of the jet, as shown. Taking ρ = 1000 kg/m³, the force exerted by the jet on the plate (in N, round off to the nearest integer) is _____



Q11 MCQ

A spherical thermocouple junction of 1.2 mm diameter (ρ = 8900 kg/m³, c = 390 J/kg·K, k = 25 W/m·K) is placed in a gas stream with h = 300 W/m²·K. Its thermal time constant is about

(A) 13.9 s
(B) 4.63 s
(C) 1.16 s
(D) 2.31 s

Q12 NAT

Wet steam at 10 bar has a dryness fraction of 0.9. At 10 bar, h_f = 762.8 kJ/kg and h_{fg} = 2015.3 kJ/kg. The specific enthalpy of the steam (in kJ/kg, round off to one decimal place) is _____

Q13 MCQ

In a simple Rankine cycle, the steam enters the turbine with h = 3230 kJ/kg and leaves it with h = 2170 kJ/kg. Saturated water leaves the condenser with h = 192 kJ/kg, and the pump work is 4 kJ/kg. The thermal efficiency of the cycle is about

(A) 32.8 %
(B) 34.8 %
(C) 37.0 %
(D) 65.2 %

Q14 NAT

In orthogonal cutting with a tool of 12° rake angle, the chip thickness ratio (uncut thickness/chip thickness) is 0.38. The shear angle (in degrees, round off to one decimal place) is _____

Q15 MCQ

Maximise Z = 3x + 2y subject to x + y ≤ 5, x + 3y ≤ 9, x ≤ 4 and x, y ≥ 0. The optimum is

(A) Z = 14 at (4, 1)
(B) Z = 13 at (3, 2)
(C) Z = 12 at (4, 0)
(D) Z = 15 at (5, 0)

Answer Key

Q	Type	Answer
Q1	NAT	20
Q2	MCQ	(C) Each of the test specimens has been numbered.
Q3	MCQ	(D) e^6
Q4	NAT	0.156 (accept 0.155 to 0.157)
Q5	MCQ	(B) 18.0 kN compression and 10.0 kN tension
Q6	NAT	30
Q7	MCQ	(A) 160 rpm in the same direction as the input
Q8	NAT	19.08 (accept 18.90 to 19.26)
Q9	MCQ	(C) 54 % of the original
Q10	NAT	636 (accept 630 to 642)
Q11	MCQ	(D) 2.31 s
Q12	NAT	2576.6 (accept 2570.0 to 2583.0)
Q13	MCQ	(B) 34.8 %
Q14	NAT	22.0 (accept 21.8 to 22.2)
Q15	MCQ	(A) Z = 14 at (4, 1)

Detailed Solutions

Q1 **Answer: 20**

Concept: Write the ages as 5x and 7x; the same number of years is added to both.

$(5x + 8)/(7x + 8) = 7/9 \rightarrow 45x + 72 = 49x + 56 \rightarrow x = 4$.

$A = 5 \times 4 =$ **20 years** (B = 28; check: 28/36 = 7/9).

Exam tip: Substitute the answer back into both ratios — it takes ten seconds and catches sign slips.

Q2 **Answer: (C) Each of the test specimens has been numbered.**

Concept: 'Each of …', 'neither of …' and 'the number of …' take a singular verb; the plural noun after 'of' does not decide the verb.

(A) and (B) wrongly use the plural 'have' / 'are'; (D) needs 'has increased' because the subject is 'the number'.

Only (C) uses the singular 'has': **(C)**.

Exam tip: 'A number of parts *have*' is plural, but 'the number of parts *has*' is singular — a favourite GATE trap.

Q3 **Answer: (D) e^6**

Concept: For a 1^∞ form, $\lim (1 + a/x)^{bx} = e^{ab}$.

$\ln L = \lim 2x \ln(1 + 3/x) = \lim 2x \times (3/x) = 6$.

$L = \mathbf{e^6}$, option (D).

Exam tip: Answer 1 is the classic trap: 1^∞ is an indeterminate form, not 1.

Q4 **Answer: 0.156 (accept 0.155 to 0.157)**

Concept: First fix k from $\int f(x)\,dx = 1$, then integrate f over the required range.

$\int_0^2 (2x - x^2)\,dx = 4 - 8/3 = 4/3$, so $k = 3/4$.

$P(X < 0.5) = (3/4)[x^2 - x^3/3]$ from 0 to 0.5 $= (3/4)(0.25 - 0.04167) =$ **0.156**.

Exam tip: The density is symmetric about $x = 1$, so $P(X > 1.5)$ is also 0.156 — a quick check.

Q5 **Answer: (B) 18.0 kN compression and 10.0 kN tension**

Concept: Method of joints — find the reactions, then take a joint with only two unknown members.

By symmetry $R_A = R_B = 15$ kN. Member AC rises 3 m over 2 m, so $\sin\theta = 3/\sqrt{13} = 0.832$ and $\cos\theta = 2/\sqrt{13} = 0.555$.

Joint A: $\Sigma F_y = 0 \rightarrow F_{AC} \sin\theta = 15 \rightarrow F_{AC} = 18.0$ kN (pushing on A: compression). $\Sigma F_x = 0 \rightarrow F_{AB} = 18.0 \times 0.555 = 10.0$ kN (tension).

Answer: **(B) 18.0 kN compression and 10.0 kN tension**.

Exam tip: In a simple roof truss the rafters are always in compression and the tie (bottom chord) in tension — use this to reject options instantly.

Q6 **Answer: 30**

Concept: $\sigma_{max} = M_{max}/Z$, with $Z = bd^2/6$; in a cantilever M_{max} is at the fixed end.

$M_{max} = wL^2/2 + PL = 6 \times 2^2/2 + 4 \times 2 = 12 + 8 = 20$ kN·m.

$Z = 100 \times 200^2/6 = 6.667 \times 10^5$ mm³, so $\sigma_{max} = 20 \times 10^6/6.667 \times 10^5 =$ **30 MPa**.

Exam tip: Keep N and mm throughout (N·mm and mm³ give MPa directly).

Q7 **Answer: (A) 160 rpm in the same direction as the input**

Concept: Train value = product of driver teeth/product of driven teeth; each external mesh reverses the direction.

$N_D = 1440 \times (20/60) \times (25/75) = 1440/9 = 160$ rpm.

Two external meshes reverse the direction twice, so the output turns the same way as the input: **(A)**.

Exam tip: Gears B and C on one shaft share a speed — never include both in the ratio as a 'driven' pair.

Q8 **Answer: 19.08 (accept 18.90 to 19.26)**

Concept: $\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$, with $\omega_n = \sqrt{(k/m)}$ and $\zeta = c/(2\sqrt{(km)})$.

$\omega_n = \sqrt{(1600/4)} = 20$ rad/s; $c_c = 2\sqrt{(1600 \times 4)} = 160$ N·s/m, so $\zeta = 48/160 = 0.3$.

$\omega_d = 20\sqrt{(1 - 0.09)} = 20 \times 0.9539 =$ **19.08 rad/s**.

Exam tip: Even 30 % damping lowers the frequency by less than 5 % — if your ω_d drops a lot, recheck ζ.

Q9 **Answer: (C) 54 % of the original**

Concept: $L_{10} = (C/P)^k$, with k = 3 for ball bearings and k = 10/3 for roller bearings.

New life/old life = $(1/1.2)^{10/3} = 0.8333^{3.333} = 0.545$.

So the life falls to about **54 %**, option (C). (58 % uses k = 3, the ball-bearing exponent.)

Exam tip: Check the bearing type before choosing the exponent — it is the only difference between the two answers.

Q10 **Answer: 636 (accept 630 to 642)**

Concept: For a single moving plate, the mass striking it per second is $\rho A(V - u)$ and the relative velocity destroyed is $(V - u)$, so $F = \rho A(V - u)^2$.

$A = \pi \times 0.06^2/4 = 2.827 \times 10^{-3}$ m²; $V - u = 15$ m/s.

$F = 1000 \times 2.827 \times 10^{-3} \times 15^2 =$ **636 N**.

Exam tip: For a series of plates (a wheel) the whole jet ρAV strikes, so $F = \rho AV(V - u)$ instead.

Q11 **Answer: (D) 2.31 s**

Concept: For lumped behaviour $\tau = \rho cV/(hA)$, and V/A = D/6 for a sphere.

Check: $Bi = h(D/6)/k = 300 \times 0.0002/25 = 0.0024 < 0.1$, so lumped analysis is valid.

$\tau = 8900 \times 390 \times 0.0002/300 =$ **2.31 s**, option (D).

Exam tip: V/A is D/6 for a sphere, D/4 for a long cylinder and half-thickness for a plate — using D itself gives the 13.9 s trap.

Q12 **Answer: 2576.6 (accept 2570.0 to 2583.0)**

Concept: For a wet mixture $h = h_f + x\,h_{fg}$.

$h = 762.8 + 0.9 \times 2015.3 = 762.8 + 1813.8$.

$h =$ **2576.6 kJ/kg**.

Exam tip: Use h_{fg}, not h_g, with the dryness fraction: $x\,h_g$ would add h_f twice.

Q13 **Answer: (B) 34.8 %**

Concept: $\eta = (W_T - W_P)/Q_{in}$, with $Q_{in} = h_1 - h_4$ and $h_4 = h_3 + W_P$.

$W_T = 3230 - 2170 = 1060$ kJ/kg; $W_{net} = 1060 - 4 = 1056$ kJ/kg.

$Q_{in} = 3230 - (192 + 4) = 3034$ kJ/kg, so $\eta = 1056/3034 =$ **34.8 %**, option (B).

Exam tip: 65.2 % is the fraction of heat rejected $(1 - \eta)$ — read what the question asks.

Q14 **Answer: 22.0 (accept 21.8 to 22.2)**

Concept: $\tan\varphi = r\cos\alpha/(1 - r\sin\alpha)$, where r is the chip thickness ratio and α the rake angle.

$r\cos\alpha = 0.38 \times 0.9781 = 0.3717$; $1 - r\sin\alpha = 1 - 0.38 \times 0.2079 = 0.9210$.

$\tan\varphi = 0.4036$, so $\varphi = \mathbf{22.0°}$.

Exam tip: A thicker chip (smaller r) means a smaller shear angle, a longer shear plane and a larger cutting force.

Q15 **Answer: (A) Z = 14 at (4, 1)**

Concept: The optimum of an LP lies at a corner of the feasible region; evaluate Z at every corner.

Corners: (0, 0), (4, 0), (4, 1) [$x = 4$ with $x + y = 5$], (3, 2) [$x + y = 5$ with $x + 3y = 9$] and (0, 3).

Z = 0, 12, 14, 13 and 6, so the maximum is **Z = 14 at (4, 1)**. (5, 0) violates $x \le 4$.

Exam tip: Always test that an option's point is feasible before comparing Z values.

What next?

Day 78 test tonight, 7 to 11 PM

65 questions · 180 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 79/90: Full-Length Mock 3. Comment **GATE90** on our reel to get all links.
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